

Design Patterns

Human Computer Interaction

Generating Design Solutions

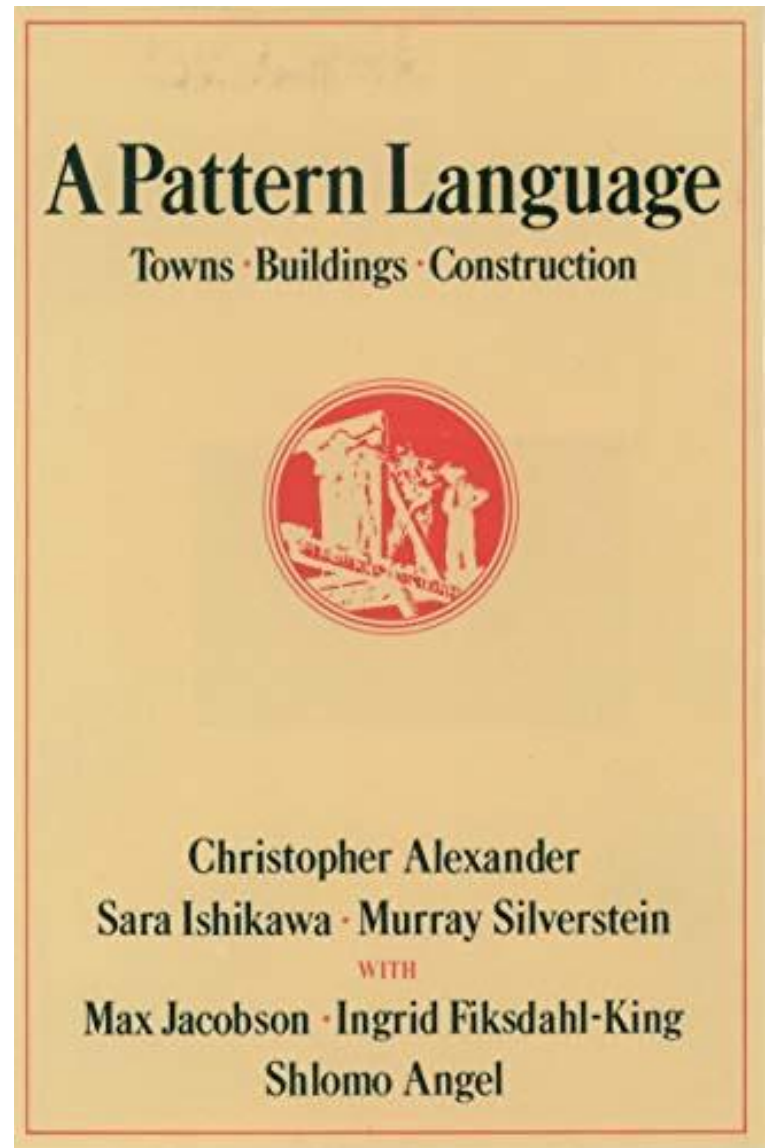
- **Design Patterns:** well-proven solutions that solve commonly recurring problems
 - they suggest a specific solution for the specific problem;
 - the solution has been tested by others;
 - the solution can be reused.

Design = Solutions

- Design is about solution
- Unfortunately, designers often reinvent things, so that it is hard to know:
 - **how** things were done before
 - **why** things were done in a certain way
 - **how** to reuse solutions

Design Patterns

- Design patterns were first used in architecture
- The idea was introduced by the architect Christopher Alexander, who defined a design pattern in this way: *"Each pattern describes a problem that occurs over and over again in our environment, and then describes the core of the solution to that problem, in such a way that you can use this solution a million times over, without ever doing it the same way twice."*



Design Patterns

- They are a way to communicate common **design problems** and **related solutions**
 - Design patterns are recurring solutions that solve common design problems
- They not too general and not too specific
 - use a solution “a million times over, without ever doing it the same way twice”
- They are a *shared language*
 - Design patterns are standard reference points for designers
 - They allow for debate over alternatives, where merely mentioning the name of a design pattern implicitly carries much more meaning than merely the name
 - They are readable by non-experts

Design Patterns As a New Literary Form

- *A literary form* is an agreement between the writer and the reader
 - A letter always starts with “Dear ...” and ends with some letter closings (e.g., “Best regards”)
- We can think about design patterns as a new literary form
 - They define certain **things** to be in a certain **place** with a certain **meaning**

Design Patterns Description

243 SITTING WALL**



1124

... if all is well, the outdoor areas are largely made up of positive spaces—POSITIVE OUTDOOR SPACES (106); in some fashion you have marked boundaries between gardens and streets, between terraces and gardens, between outdoor rooms and terraces, between play areas and gardens—GREEN STREETS (51), PEDESTRIAN STREET (100), HALF-HIDDEN GARDEN (111), HIERARCHY OF OPEN SPACE (114), PATH SHAPE (121), ACTIVITY POCKETS (124), PRIVATE TERRACE ON THE STREET (140), OUTDOOR ROOM (163), OPENING TO THE STREET (165), GALLERY SURROUND (166), GARDEN GROWING WILD (172). With this pattern, you can help these natural boundaries take on their proper character, by building walls, just low enough to sit on, and high enough to mark the boundaries.

If you have also marked the places where it makes sense to build seats—SEAT SPOTS (241), FRONT DOOR BENCH (242)—you can kill two birds with one stone by using the walls as seats which help enclose the outdoor space wherever its positive character is weakest.

In many places walls and fences between outdoor spaces are too high; but no boundary at all does injustice to the subtlety of the divisions between the spaces.

Consider, for example, a garden on a quiet street. At least somewhere along the edge between the two there is a need for a seam, a place which unites the two, but does so without breaking down the fact that they are separate places. If there is a high wall or a hedge, then the people in the garden have no way of being connected to the street; the people in the street have no way of being connected to the garden. But if there is no barrier at all—then the division between the two is hard to maintain. Stray dogs can wander in and out at will; it is even uncomfortable to sit in the garden, because it is essentially like sitting in the street.

1125

CONSTRUCTION

The problem can only be solved by a kind of barrier which functions as a barrier which separates, and as a seam which joins, at the same time.

A low wall or balustrade, just at the right height for sitting, is perfect. It creates a barrier which separates. But because it invites people to sit on it—invites them to sit first with their legs on one side, then with their legs on top, then to swivel round still further to the other side, or to sit astride it—it also functions as a seam, which makes a positive connection between the two places.

Examples: A low wall with the children's sandbox on one side, circulation path on the other; low wall at the front of the garden, connecting the house to the public path; a sitting wall that is a retaining wall, with plants on one side, where people can sit close to the flowers and eat their lunch.

Ruskin describes a sitting wall he experienced:

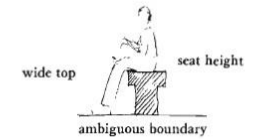
Last summer I was lodging for a little while in a cottage in the country, and in front of my low window there were, first, some beds of daisies, then a row of gooseberry and currant bushes, and then a low wall about three feet above the ground, covered with stone-crests. Outside, a corn-field, with its green ears glistening in the sun, and a field path through it, just past the garden gate. From my window I could see every peasant of the village who passed that way, with basket on arm for market, or spade on shoulder for field. When I was inclined for society, I could lean over my wall, and talk to anybody; when I was inclined for science, I could botanize all along the top of my wall—there were four species of stone-crest alone growing on it; and when I was inclined for exercise, I could jump over my wall, backwards and forwards. That's the sort of fence to have in a Christian country; not a thing which you can't walk inside of without making yourself look like a wild beast, nor look at out of your window in the morning without expecting to see somebody impaled upon it in the night. (John Ruskin, *The Two Paths*, New York: Everyman's Library, 1907, p. 203.)

Therefore:

Surround any natural outdoor area, and make minor boundaries between outdoor areas with low walls, about 16 inches high, and wide enough to sit on, at least 12 inches wide.

1126

243 SITTING WALL



Place the walls to coincide with natural seat spots, so that extra benches are not necessary—SEAT SPOTS (241); make them of brick or tile, if possible—SOFT TILE AND BRICK (248); if they separate two areas of slightly different height, pierce them with holes to make them balustrades—ORNAMENT (249). Where they are in the sun, and can be large enough, plant flowers in them or against them—RAISED FLOWERS (245).

1127

Design Patterns Description

Name

243 SITTING WALL**

Images



Context

... if all is well, the outdoor space is a mix of positive spaces—POSITIVE OUTDOOR SPACE (100); in some fashion you have marked boundaries between gardens and streets, between terraces and gardens, between outdoor rooms and terraces, between play areas and gardens—GREEN STREETS (51), PEDESTRIAN STREET (100), HALF-HIDDEN GARDEN (111), HIERARCHY OF OPEN SPACE (114), PATH SHAPE (121), ACTIVITY POCKETS (124), PRIVATE TERRACE ON THE STREET (140), OUTDOOR ROOM (163), OPENING TO THE STREET (165), GALLERY SURROUND (166), GARDEN GROWING WILD (172). With this pattern, you can help these natural boundaries take on their proper character, by building walls, just low enough to sit on, and high enough to mark the boundaries.

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In many places walls and fences between outdoor spaces are too high; but no boundary at all does injustice to the subtlety of the divisions between the spaces.

Consider, for example, a garden on a quiet street. At least somewhere along the edge between the two there is a need for a seam, a place which unites the two, but does so without breaking down the fact that they are separate places. If there is a high wall or a hedge, then the people in the garden have no way of being connected to the street; the people in the street have no way of being connected to the garden. But if there is no barrier at all—then the division between the two is hard to maintain. Stray dogs can wander in and out at will; it is even uncomfortable to sit in the garden, because it is essentially like sitting in the street.

CONSTRUCTION

The problem can only be solved by a kind of barrier which functions as a barrier which separates, and as a seam which joins, at the same time.

A low wall or balustrade, just at the right height for sitting, is perfect. It creates a barrier which separates. But because it invites people to sit on it—invites them to sit first with their legs on one side, then with their legs on the other—it is still further to the other side, as a seam, which makes places.

Examples: A low wall with the children's sandbox on one side, circulation path on the other; low wall at the front of the garden, connecting the house to the public path; a sitting wall that is a retaining wall, with plants on one side, where people can sit close to the flowers and eat their lunch.

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Then

Surround any natural outdoor area, and make minor boundaries between outdoor areas with low walls, about 16 inches high, and wide enough to sit on, at least 12 inches wide.

243 SITTING WALL



References to other patterns

Place the walls to coincide with natural seat spots, so that extra benches are not necessary—SEAT SPOTS (241); make them of brick or tile, if possible—SOFT TILE AND BRICK (248); if they separate two areas of slightly different height, pierce them with holes to make them balustrades—ORNAMENT (249). Where they are in the sun, and can be large enough, plant flowers in them or against them—RAISED FLOWERS (245). . . .

Design Patterns Balance Forces

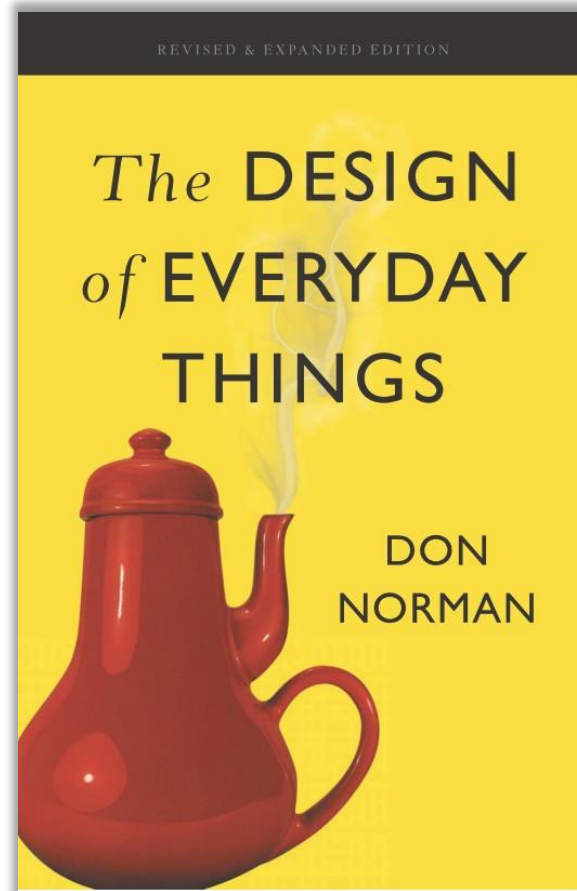
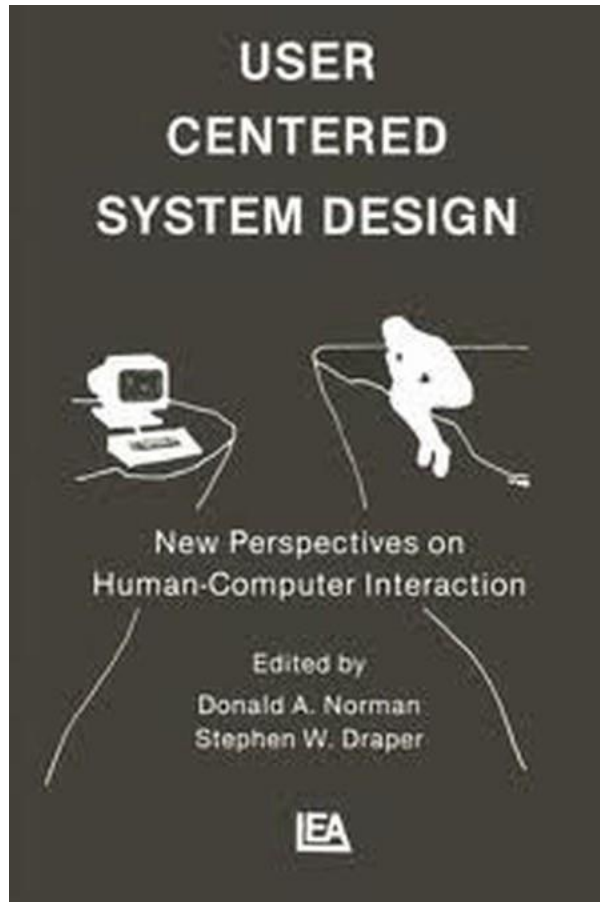
- Design patterns solve a problem **of conflicting forces**
- Example:
 - People are naturally drawn towards light
 - But like to sit

Design Patterns Balance Forces

- Design patterns solve a problem of **conflicting forces**
- Example:
 - People are naturally drawn towards light
 - But like to sit
- Solution:
 - Alexander's **Window Seat** pattern



Patterns in HCI



UI Design Patterns

- Each user interface is unique and has its own set of goals and data, but...
 - This does not mean that we should force users to learn new conventions to operate them!
- With UI design patterns, we can accelerate our users understanding of the interface

UI Design Patterns

Design patterns

User Interface Design patterns are recurring solutions that solve common design problems. Design patterns are standard reference points for the experienced user interface designer.

Dwell into the patterns below to learn a common language of web design.

Design patterns provide a common language between designers. They allow for debate over alternatives, where merely mentioning the name of a design pattern implicitly carries much more meaning than merely the name.

User Interface Design Patterns

Getting input	Navigation	Dealing with data	Social
Forms - Password Strength Meter - Structured Format - Captcha - Rule Builder - Keyboard Shortcuts - Drag and drop - Inplace Editor - Preview - Expandable Input - Autosave - Input Prompt - Good Defaults - Fill in the Blanks - WYSIWYG - Input Feedback	Tabs - Navigation Tabs - Module Tabs Jumping in hierarchy - Breadcrumbs - Fat Footer - Notifications - Modal - Home Link - Shortcut Dropdown Menus - Vertical Dropdown Menu - Accordion Menu - Horizontal Dropdown Menu Content	Tables - Alternating Row Colors - Sort By Column - Table Filter Formatting data - Frequently Asked Questions (FAQ) - Dashboard - Copy Box Images - Gallery - Slideshow - Image Zoom Search - Autocomplete - Search Filters	Reputation - Collectible Achievements - Testimonials - Leaderboard Social interactions - Activity Stream - Auto-sharing Mini - Friend list Mini - Reaction - Chat - Follow - Invite friends - Friend

<https://ui-patterns.com/patterns>

UI Design Patterns

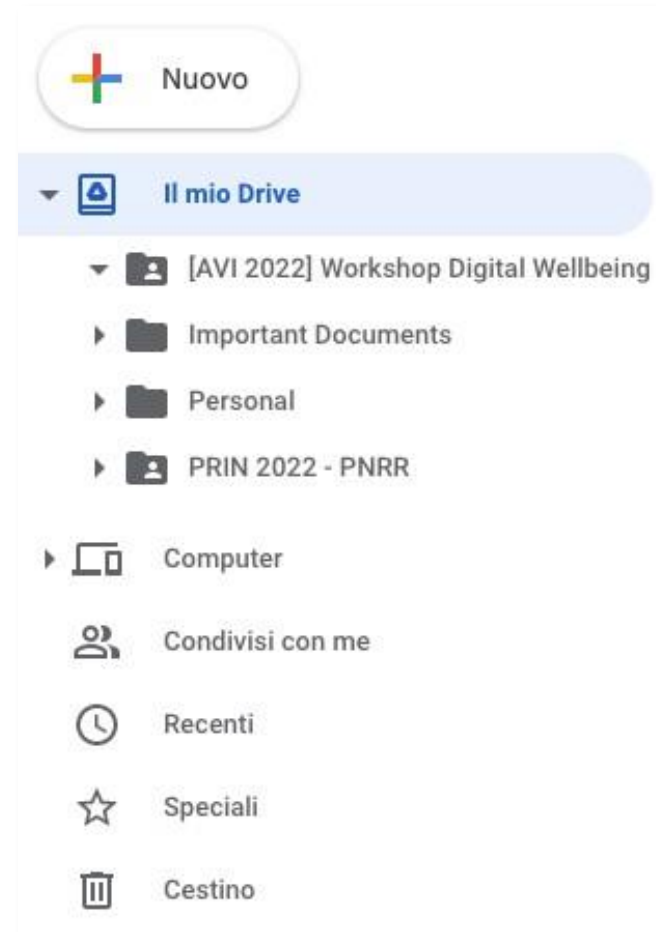
ACCORDION MENU

DROPDOWN MENU

CARDS

BREADCRUMBS

THE HAMBURGER



UI Design Patterns

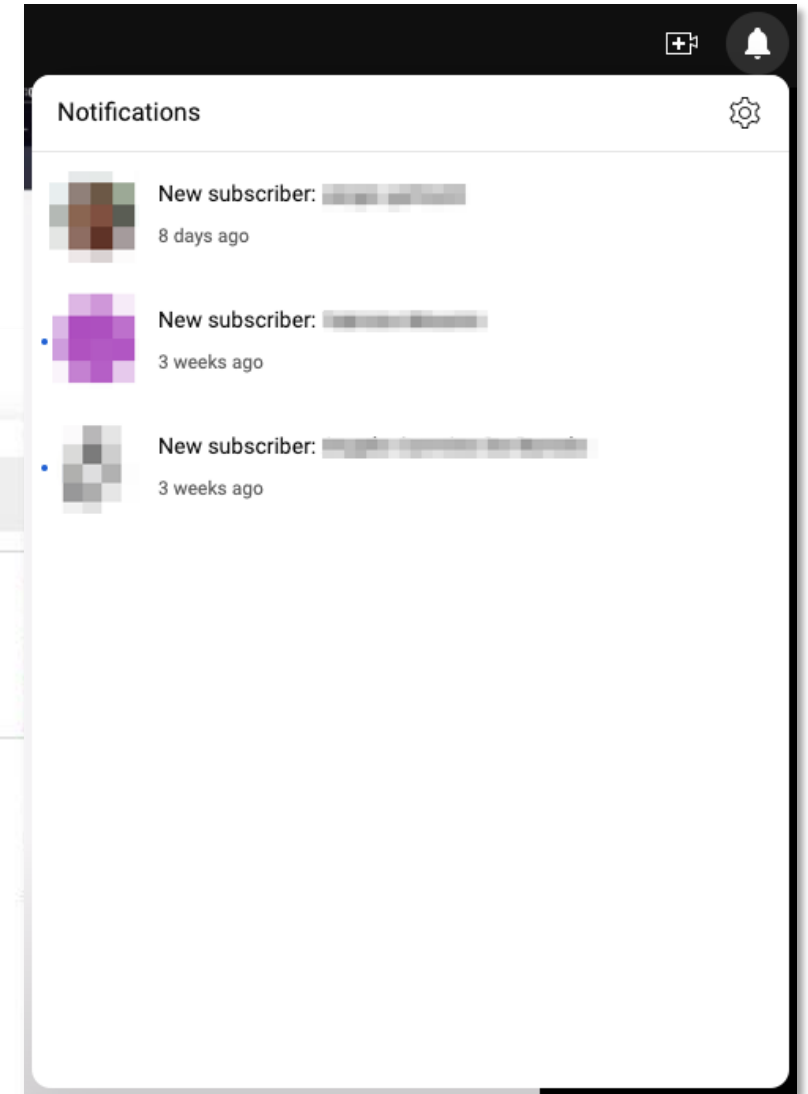
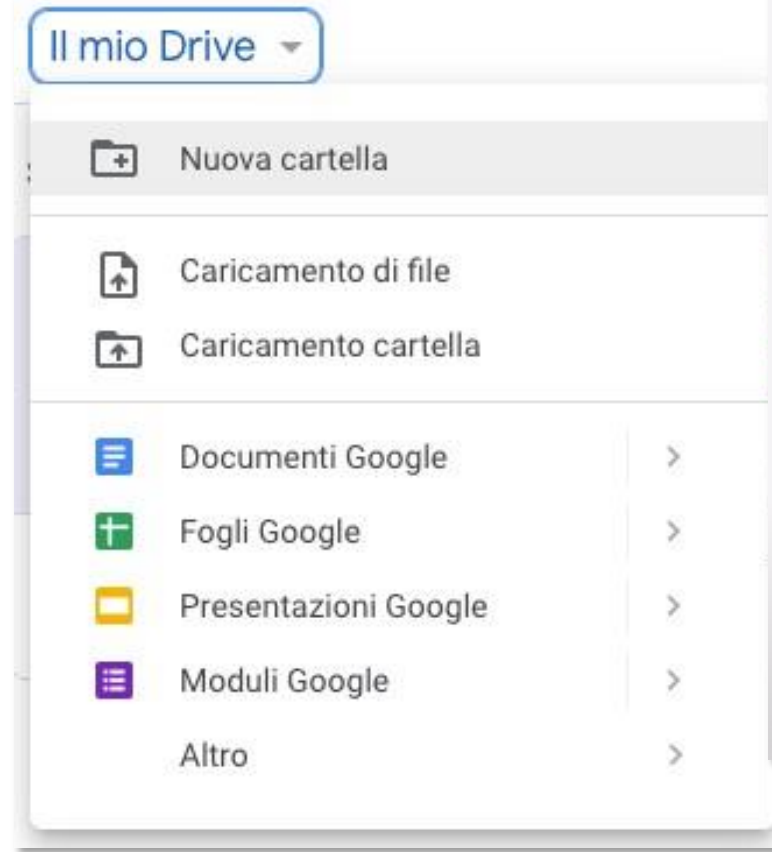
ACCORDION MENU

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THE HAMBURGER

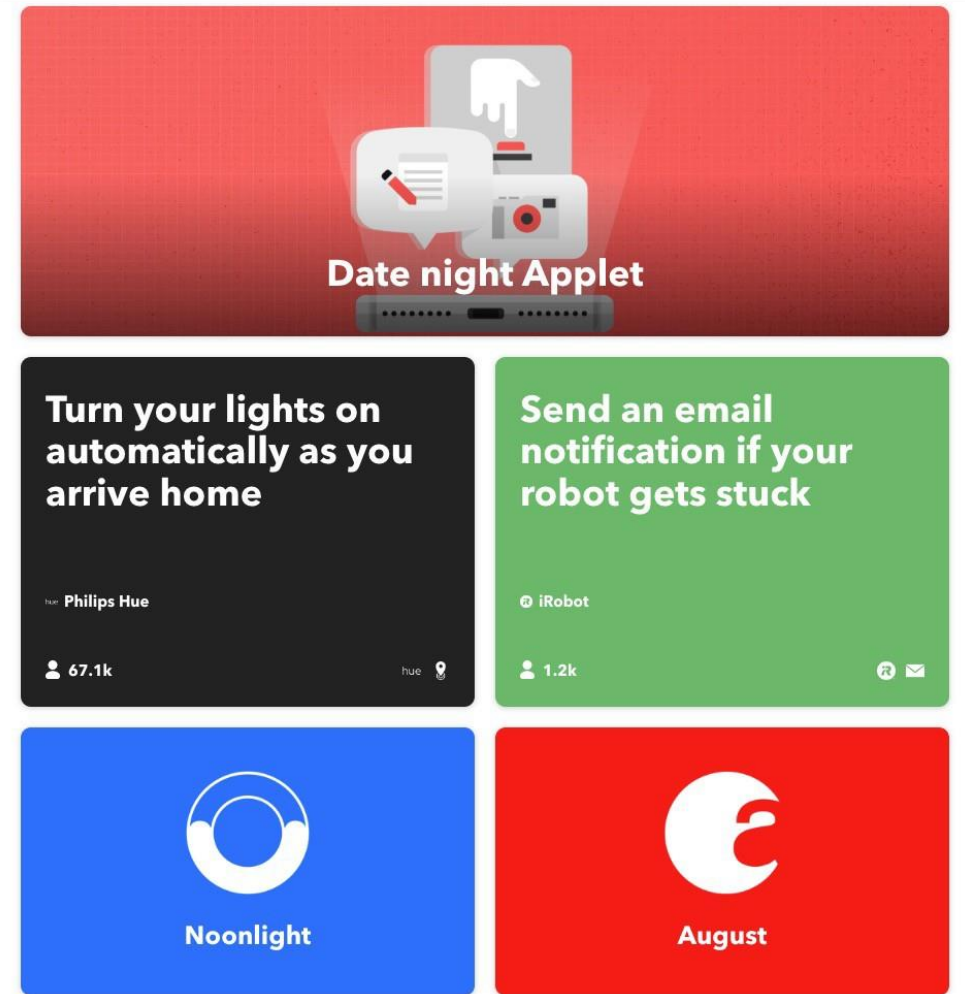


UI Design Patterns

CARDS

Use visually distinct containers (cards) to present content in a clean, organized, and easily digestible manner.

Each card typically contains a specific piece of information or a set of related items, such as text, images, links, and interactive elements.



UI Design Patterns

BREADCRUMBS

Use linked labels to provide secondary navigation that shows **the path from the front to the current site page in the hierarchy**.

[Cart](#) > [Billing](#) > [Shipping](#) > **Payment**



[Cart](#) / [Billing](#) / [Shipping](#) / **Payment**



UI Design Patterns

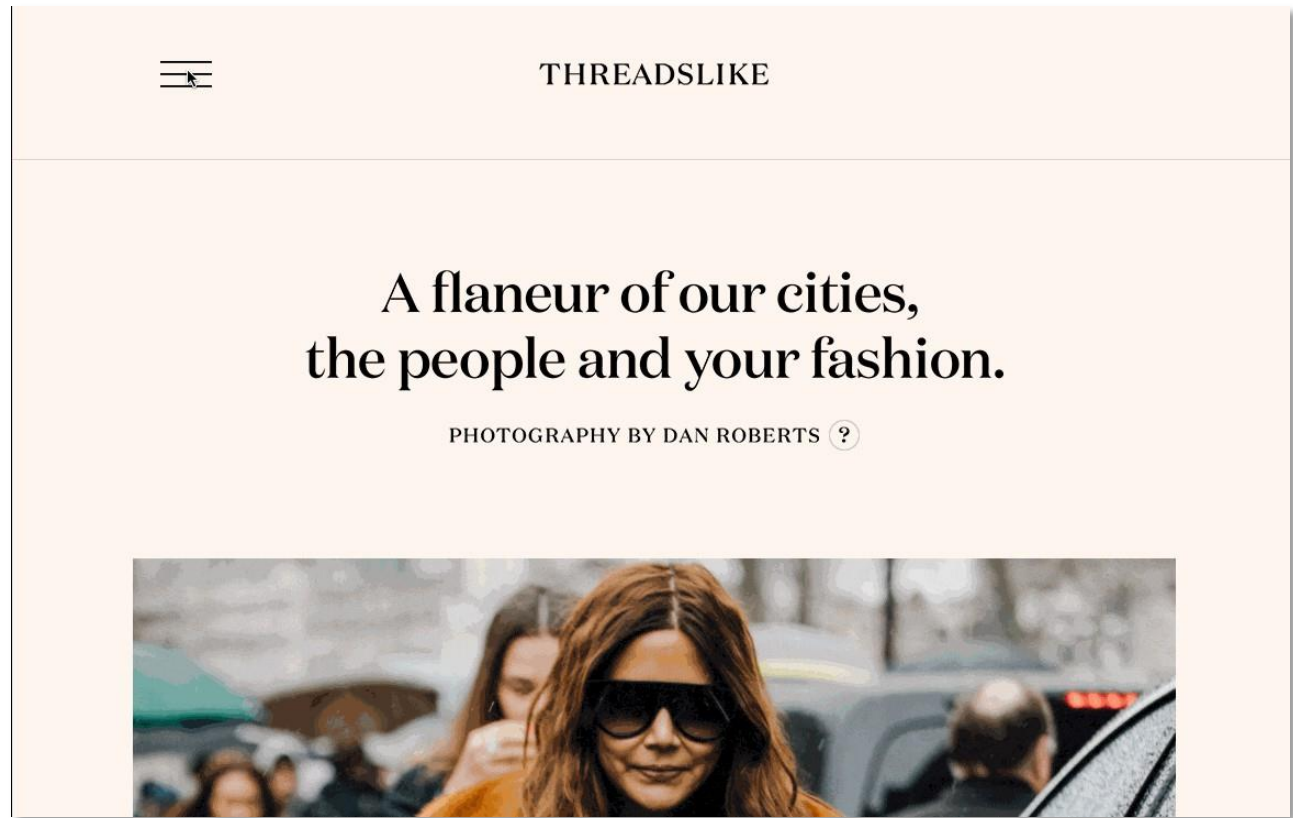
ACCORDION MENU

DROPDOWN MENU

CARDS

BREADCRUMBS

THE HAMBURGER



Design Patterns in (Android) Mobile Apps

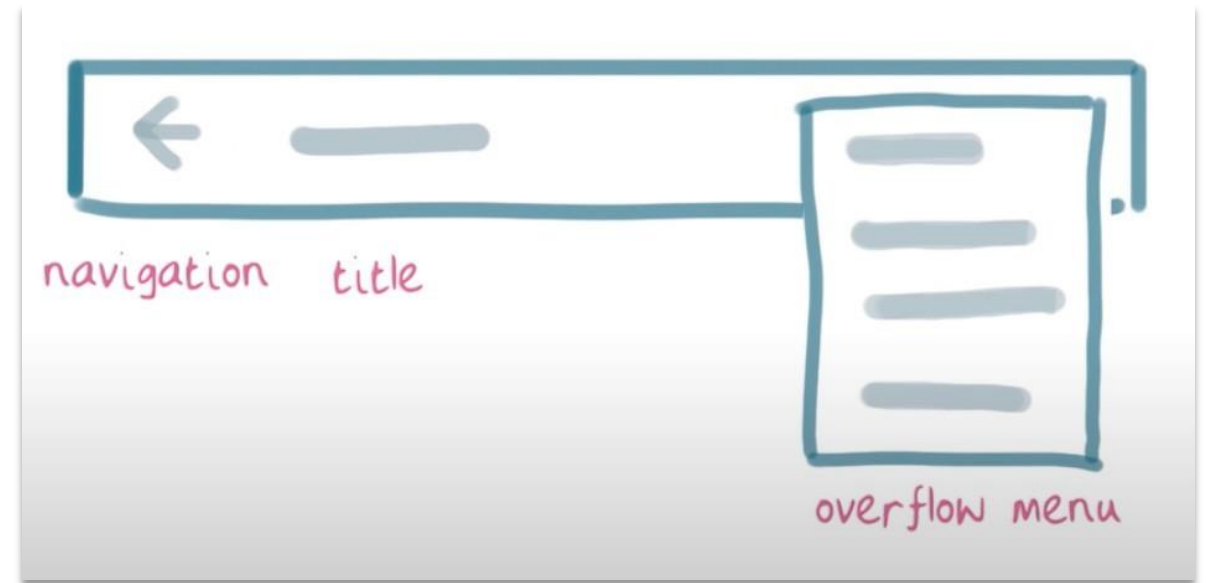
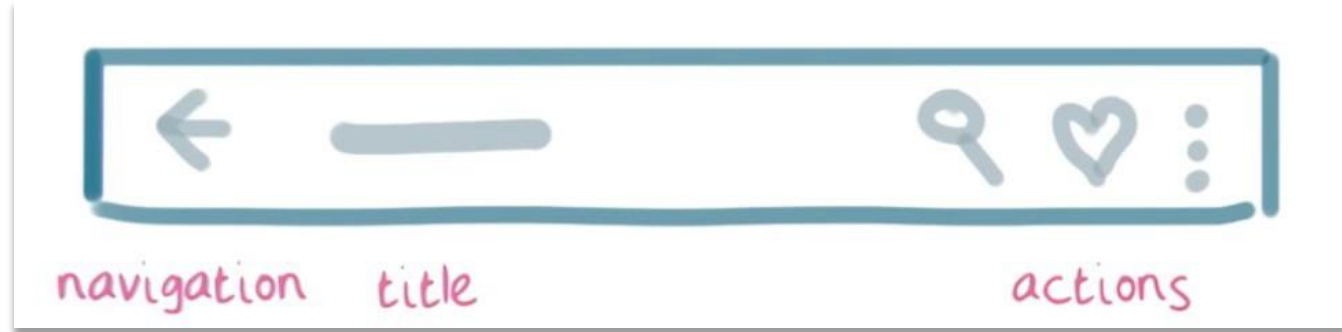
TOOLBAR

APP BAR

TABS

NAVIGATION DRAWER

SCROLLING AND PAGING



Design Patterns in (Android) Mobile Apps

TOOLBAR

APP BAR

TABS

NAVIGATION DRAWER

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Design Patterns in (Android) Mobile Apps

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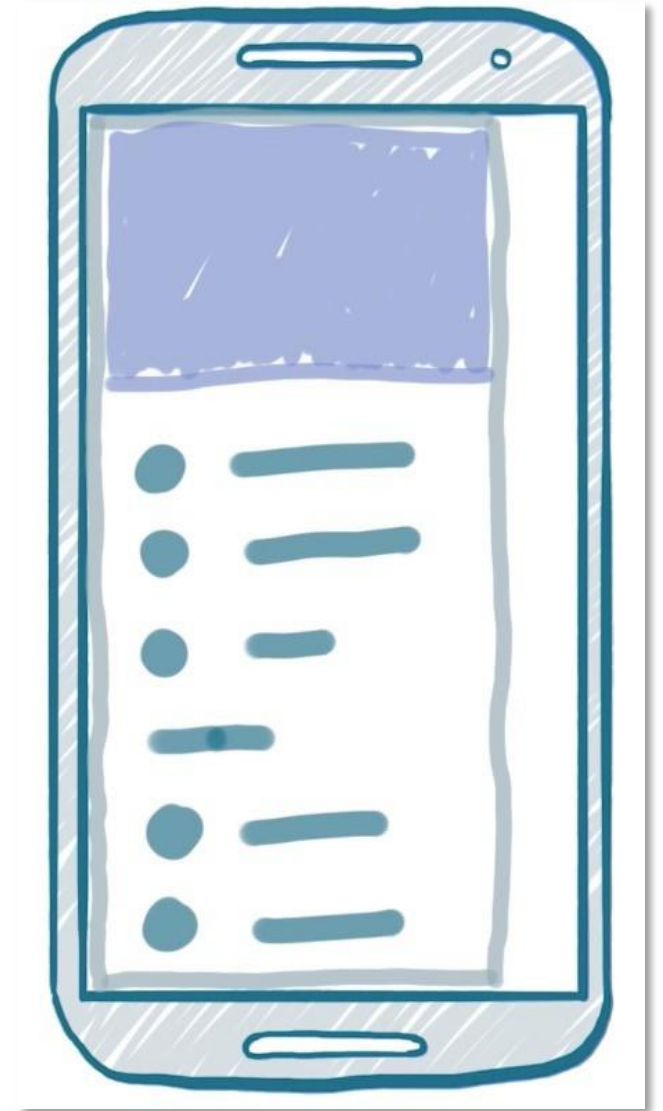
TOOLBAR

APP BAR

TABS

NAVIGATION DRAWER

SCROLLING AND PAGING



Design Patterns in (Android) Mobile Apps

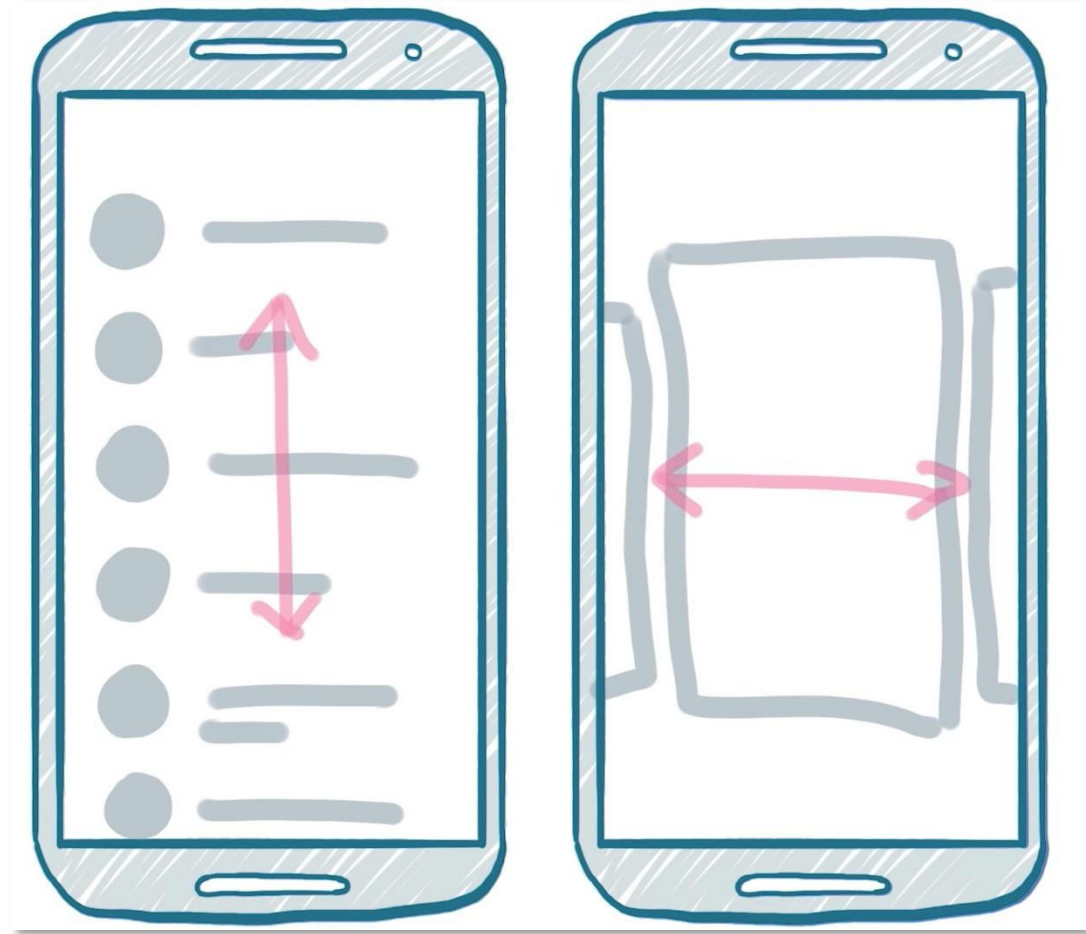
TOOLBAR

APP BAR

TABS

NAVIGATION DRAWER

SCROLLING AND PAGING



Dark Patterns

Deceptive designs that go against users' best interests

Some designers use **dark patterns** to lead or **trick users into performing certain actions**, typically in e-commerce, so they spend more or surrender personal information.

You're trying to cancel a membership, and the following prompt appears. **Is there a dark pattern on this page?**

**Are you sure you want to
cancel your membership?**

You will no longer be able to use
our services if you do.

Continue

Cancel



Dark Pattern

**Are you sure you want to
cancel your membership?**

You will no longer be able to use
our services if you do.

Continue

Cancel

Ethical Pattern

**Are you sure you want to
cancel your membership?**

You will no longer be able to use
our services if you do.

Yes, I would like to cancel my account

No, I would like to keep my account



Dark Patterns

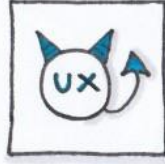
Often disguised as normal user interface elements or content, dark patterns can deceive users into:

- *Paying more for a service or product*
- *Subscribing or opting in to ongoing communications from the company*
- *Sharing their data for company use*
- *Making ill-informed decisions based on fake or vague information*
- *Agreeing to unsavory terms and conditions*

Dark Patterns

- The term “dark pattern” was coined in 2010 by **Harry Brignull**, a design practitioner
- The aim was to include all those designs that are **deliberately** adopted to promote choices that are not in the user’s best interest

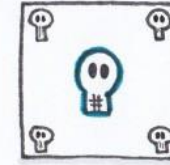
Common Dark Patterns



DARK PATTERNS

©KRISZTINA SZEROVAY
WWW.SKETCHINGFORUX.COM

UX Knowledge Base Sketch #29



DARK PATTERNS ARE

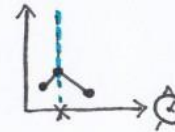


TRICKS

THAT MAKE THE USERS DO THINGS THEY DID NOT MEANTO.

FINE LINE BETWEEN
INFLUENCING USERS'
BEHAVIOR AND
TRICKING THEM!

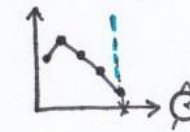
DARK PATTERNS CAN WORK
IF SOMEONE IS LOOKING FOR
SHORT-TERM RESULTS.



ONLY INITIAL
SUCCESS, NOT
SUSTAINABLE

vs.

USING DARK PATTERNS
HAS A NEGATIVE IMPACT
IN THE LONG-TERM



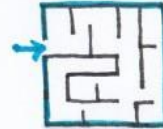
USERS WILL
SWITCH TO
MORE ETHICAL
PRODUCTS / SERVICES

THE MOST COMMON TYPES



FORCED CONTINUITY

FREE TRIAL → CHANGES
TO A PAYING SCHEME WITHOUT
WARNING



ROACH MOTEL

THE START IS EASY
(SIGN UP / SUBSCRIPTION),
QUITTING IS HARD



GROWTH HACKING THROUGH SPAMMING

YOU BECOME THE SPAMMER
WITHOUT KNOWING IT



DELIBERATE MISDIRECTION

FOCUSING THE USERS' ATTENTION
ON THE MORE EXPENSIVE OPTION,
HIDING THE CHEAPER WAY



OBSCURED PRICING

MAKING IT HARD TO
COMPARE THE PRICES



SNEAK INTO BASKET

A RANDOM ADDITIONAL ITEM
APPEARS IN YOUR BASKET
(WITHOUT YOUR CONSENT)



BAIT & SWITCH

USE A CONVENTION, PATTERN
IN A WAY TO MAKE THE USER
FALSELY ASSUME SOMETHING



DISGUISED ADS

AN AD LOOKING LIKE
ANOTHER TYPE OF CONTENT
/ NAVIGATION



ROADBLOCK

A POP-UP INTERRUPTS YOUR
INTENDED ACTION



HIDDEN COSTS

AT THE CHECKOUT, A NEW,
UNEXPECTED COST APPEARS



PRIVACY ZUCKERING

SHARING MORE PRIVATE INFO
THAN YOU WANT



MISINFORMATION

E.G. CONFUSING
- COLOR, CONTRAST
- LANGUAGE

TRICK
QUESTIONS
CHECK BOX
TREACHERY

Hidden costs

- It involves tacking additional fees (i.e., service fees, shipping costs, VAT) onto your order to make the end price significantly higher than originally advertised.

Order Subtotal	\$50.98
Standard Delivery	\$14.99
Care & Handling	\$2.99
Tax	\$4.56
Total	\$73.52
Savings Today ⓘ	\$9.00

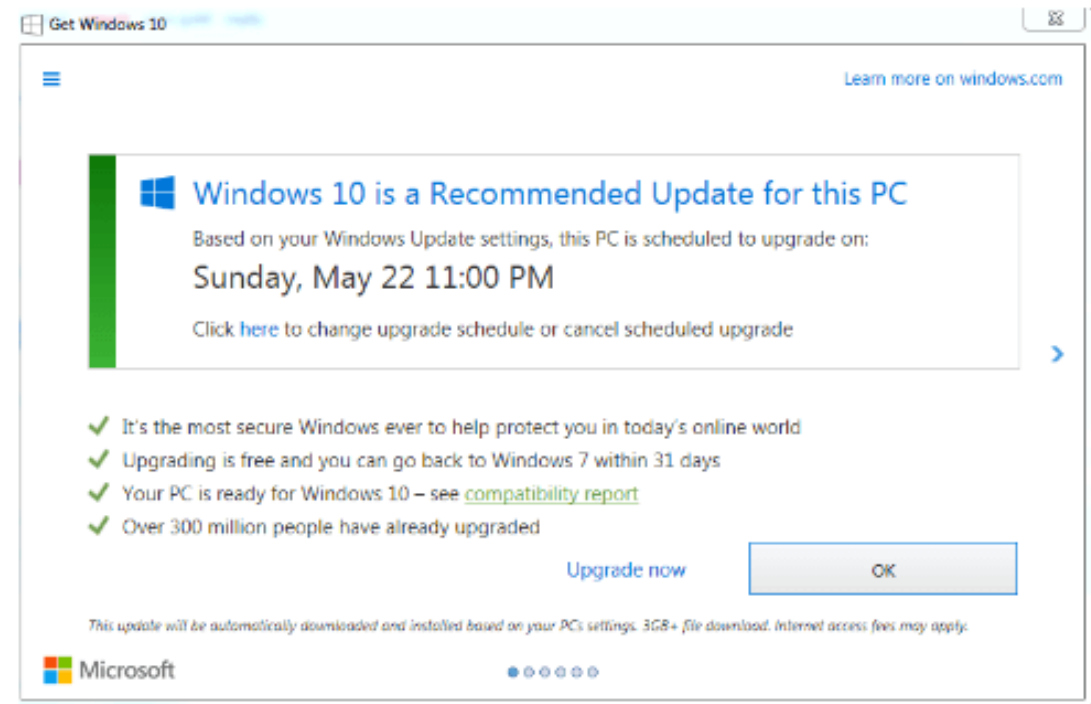
Get a Delivery Rebate up to \$15 for your Proflowers purchase! [Learn More](#)

Bait and switch

- companies ‘bait’ users into clicking on buttons they think will lead them to certain pages or initiate a specific action—only for something unexpected to happen instead.

Example:

- When Windows customers tried to close a pop-up asking them to upgrade the software—but discovered that the ‘close’ button triggered an immediate software update.



Forced continuity

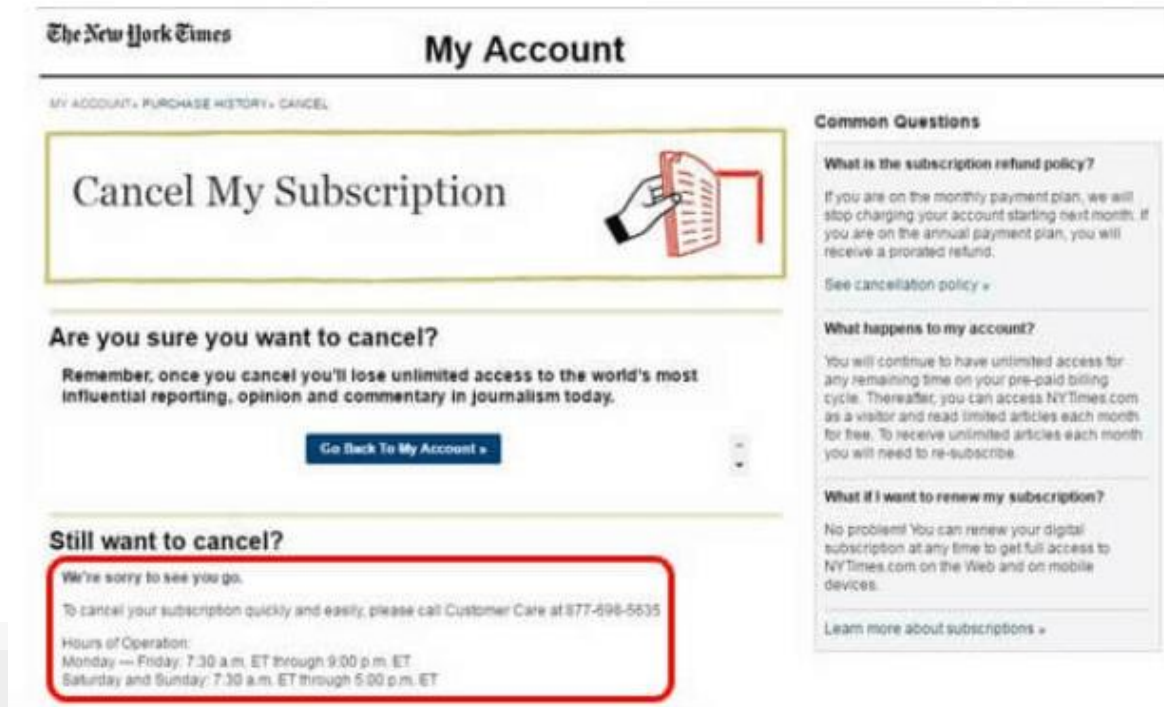
- Forced continuity takes place when [a company tricks you into entering your payment details to sign up for a 'free trial'](#), only to start billing you as soon as that free trial ends (with no warning)



Roach motel

- When companies make **signing up intentionally straightforward** (i.e., a two-step process) but
- add **unnecessary steps to cancelling your account**—or
- bury the ‘unsubscribe’ button in a small, unsuspecting corner of the site.

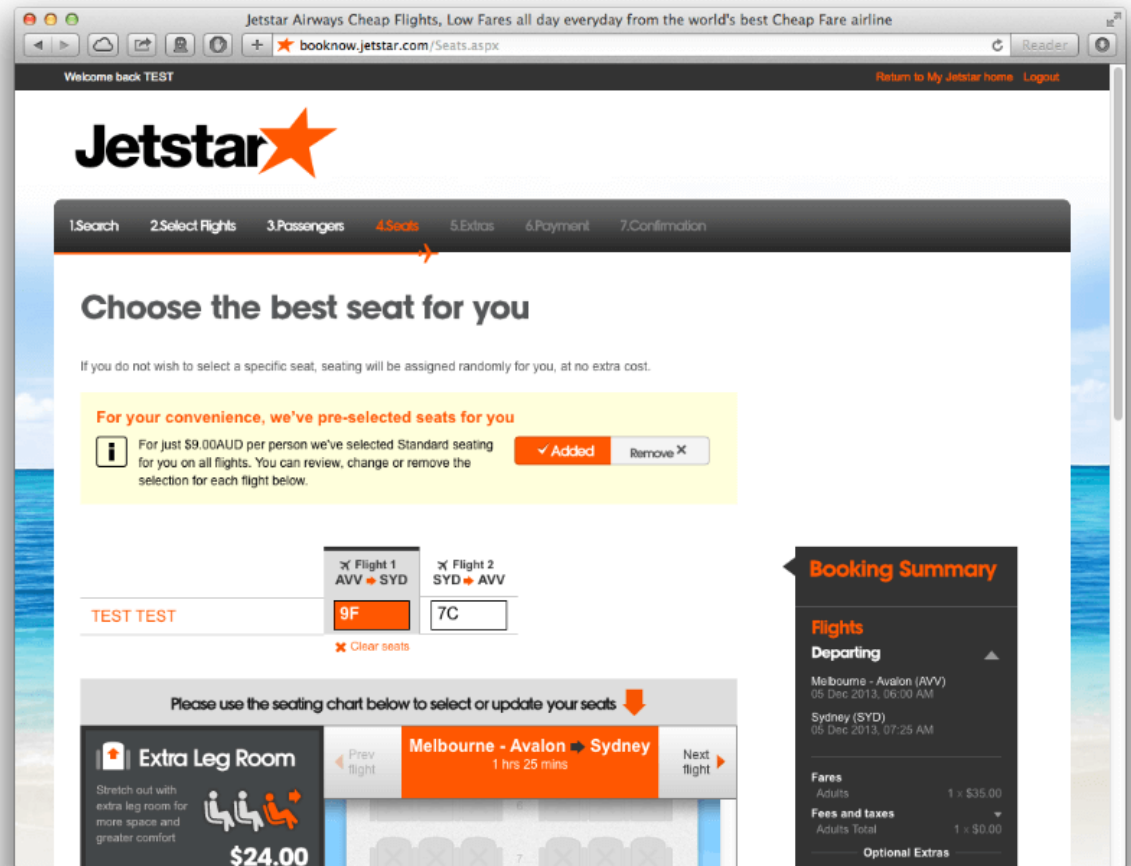
In this case, companies are banking on users getting so frustrated that they eventually give up and keep the subscription!



Misdirection

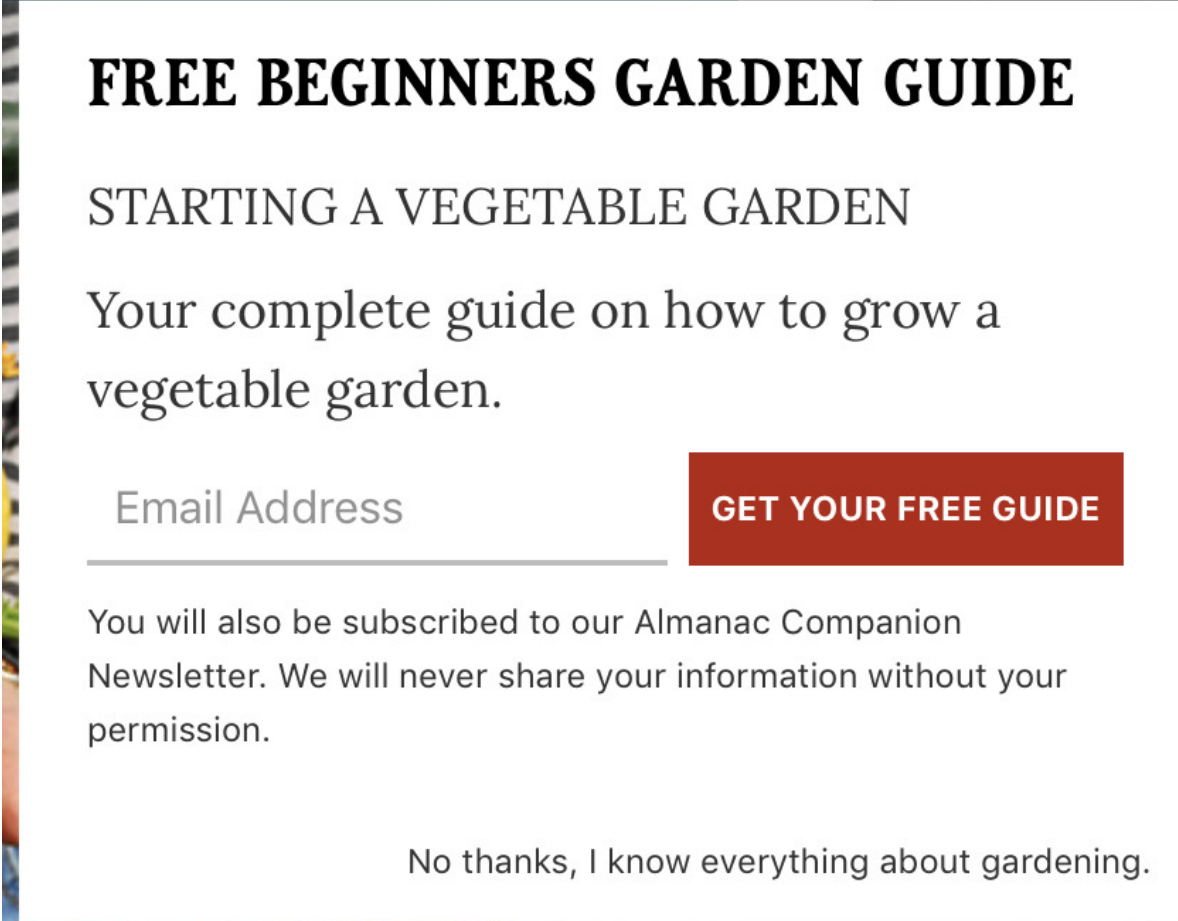
- Misdirection distracts the users from the task at hand by encouraging them to focus on something different.

In this example, this aviation company is distracting the user from selecting their own seats with a bright banner that lets you know that they've pre-selected the seats on your behalf, with an added cost.



Confirm-shaming

- Confirm-shaming uses clever wording to shame users for opting out or declining a service.
- It works by triggering uncomfortable emotions, such as guilt or shame, to influence users' decision-making.

A screenshot of a web form titled "FREE BEGINNERS GARDEN GUIDE" with the subtitle "STARTING A VEGETABLE GARDEN". The text describes it as a "Your complete guide on how to grow a vegetable garden." Below this is an input field labeled "Email Address" and a red button labeled "GET YOUR FREE GUIDE". A disclaimer states: "You will also be subscribed to our Almanac Companion Newsletter. We will never share your information without your permission." At the bottom, there is a link that says "No thanks, I know everything about gardening."

FREE BEGINNERS GARDEN GUIDE

STARTING A VEGETABLE GARDEN

Your complete guide on how to grow a vegetable garden.

Email Address

GET YOUR FREE GUIDE

You will also be subscribed to our Almanac Companion Newsletter. We will never share your information without your permission.

[No thanks, I know everything about gardening.](#)

Sneak into basket

- As the name suggests, this dark pattern sees companies sneak undesired items into users' baskets somewhere along their purchasing journey.

Friend spam

- Friend spam is when a platform asks for your social media login details under the guise of helping you connect with people you know.
- Instead, they spam your contacts with messages that appear as if they've come from you, but have actually come from the system.

Connect with your friends

Find which one of your friends joined our service!
By adding your email your friends are also able to
add you to their network!

Johndoe@company.com

Continue

*By clicking continue, you allow us to mail every contact you have ever had on your behalf.

Disguised ads

- When you visit a page, ads may be disguised as something else, like an article or news item, or placed in a way that makes them hard to distinguish from the real content, like a 'download' button on a free font website.

The screenshot shows the 'The Blacklist' font page on Dafont.com. The page features a red header bar with the text 'The Blacklist by Oodtype' and a 'Download' button. A red box highlights the 'Download' button, which is disguised as a 'Download' button for a font. The page also displays the font's name 'The Blacklist' in a large, stylized script, a 'Note of the author' stating 'THIS FONT IS FREE FOR PERSONAL USE...', and a table of the font's characters (A-Z, a-z) in the same script. On the right side, there is a 'Start Download' section with a 'Convert All Files' button and a red arrow pointing to the right.

The Blacklist by Oodtype

The Blacklist

Note of the author
... THIS FONT IS FREE FOR PERSONAL USE...
Commercial license font are available there :
oodytype@gmail.com

First seen on Dafont: November 06, 2017

A	B	C	D	E	F	G	H	I	J	K	L	M	N
1065	1066	1067	1068	1069	1070	1071	1072	1073	1074	1075	1076	1077	1078
A	B	C	D	E	F	G	H	I	J	K	L	M	N

O	P	Q	R	S	T	U	V	W	X	Y	Z
1079	1080	1081	1082	1083	1084	1085	1086	1087	1088	1089	1090
O	P	Q	R	S	T	U	V	W	X	Y	Z

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
1091	1092	1093	1094	1095	1096	1097	1098	1099	1100	1101	1102	1103	1104	1105	1106	1107	1108
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r

Start Download

Get Convert All Files for OSX. 100% Safe & Secure, Download Free!

Convert All Files

Dark Patterns

- **Harry Brignull** published a gallery of dark patterns on the www.darkpatterns.org
- He launched an hall of shame campaign on Twitter using the hashtag **#darkpatterns**

Dark Patterns ... or Deceptive Designs

- Many organizations are moving away from the oppressive terminology typical in computer science
 - For example from “master/slave” to “parent/child” and from “blacklist” to “block list”
- In the case of “dark patterns,” the association of “dark” with harm is problematic
 - it may reinforce the racist heuristic of viewing people with darker skin tones as evil (i.e., the “bad is black” effect)
- There are nowadays alternative names for dark patterns as well, such as **deceptive designs**

Attention-Capture Damaging Patterns

A particular kind of dark patterns

Attention Economy

- Why is our digital well-being undermined by contemporary technology?
- Most of the contemporary tech-companies adopt a business model that is called the “Attention Economy”
- Your attention is the most valuable commodity in modern digital economy
 - Our **attention** is transformed into a **currency**
 - We «**pay**» for a service with the time we spend on it
 - Tech companies **may sell** our attention (and our data, sometimes) to advertising companies
- This business model is convenient
 - **Alphabet** (the company that owns Google) is worth \$1 trillion
 - **Meta** (which owns Facebook, Instagram and WhatsApp) is worth about \$700 billion

Attention Economy

2020 This Is What Happens In An Internet Minute



2021 This Is What Happens In An Internet Minute



Attention-Capture Damaging Patterns: Definition

- A recurring pattern in digital interfaces that a designer **uses to** exploit psychological vulnerabilities and **capture attention**, often leading the user to lose **track of their goals**, lose their **sense of time and control**, and later **feel regret**
- The goal of ACDPs is to maximize continuous usage, daily visits, and interactions (e.g., clicks, shares, likes, etc.).
- They make users more likely to visit a digital service again and click on similar types of rewarding content, thus creating a «**trap**» for the user that enables the **stakeholder's goal**

Attention-Capture Damaging Patterns: Strategies

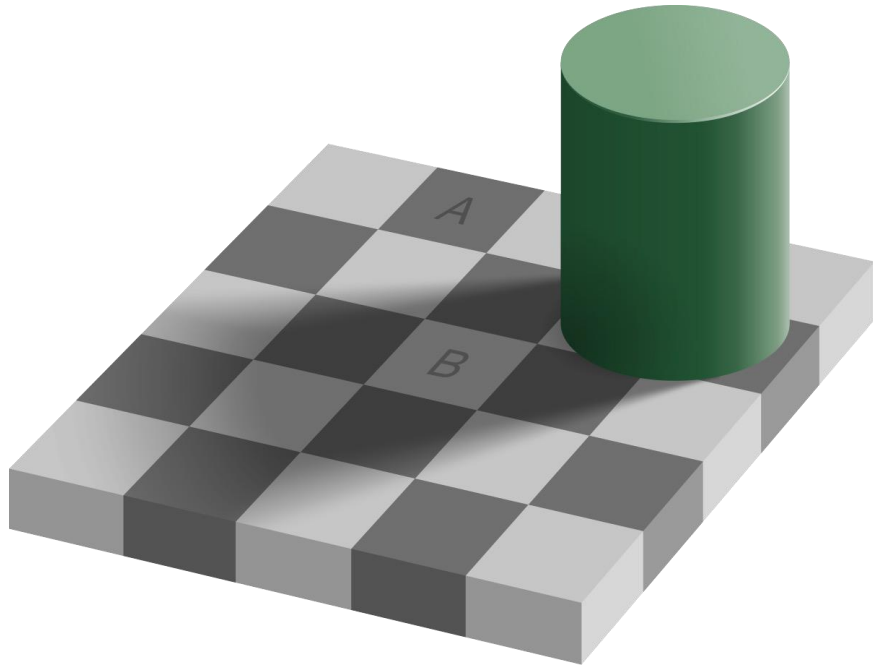
- Typically, ACDPs exploit the need for autonomous decision making by "automating" processes and functions
- Paradoxically, they can **improve the usability** of a platform
- They exploit users' **psychological vulnerabilities**:
 - Variable reward
 - Immediate gratification
 - [...]

Attention-Capture Damaging Patterns: Strategies

- ACDPs exploit **psychological vulnerabilities** and “**cognitive biases**”
- A cognitive bias is a **systematic error** in thinking that occurs when people are processing and interpreting information in the world around them and affects the decisions and judgments that they make.

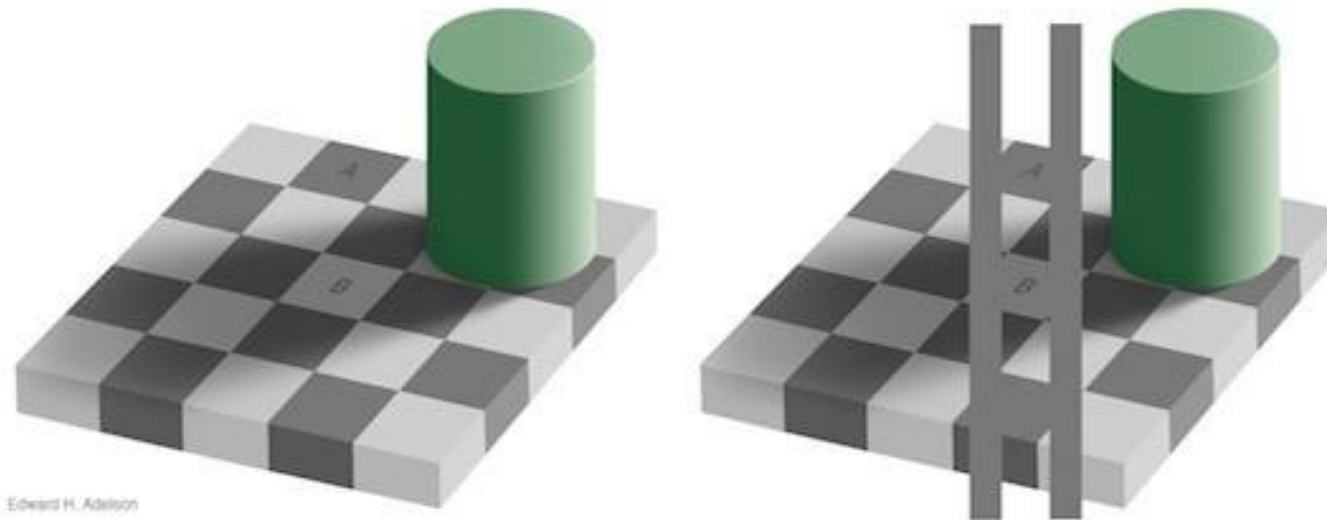
Attention-Capture Damaging Patterns: Strategies

- ACDPs exploit **psychological vulnerabilities** and “**cognitive biases**” of users
 - What we perceive as true depends on the context in which we see it!



Attention-Capture Damaging Patterns: Strategies

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Edward H. Adelson

[Checkers shadow Illusion](#)

Attention-Capture Damaging Patterns: Strategies

- The exploited psychological vulnerabilities can be of various type:
 - Notifications (with their vibrations, flashing LEDs ...) mimic danger signals that we may experience in “offline” life
 - The possibility of receiving new comments or likes keeps us in a persistent state of alert (**variable reward**)
 - Every time we receive a comment or a “like,” our brain gets a dose of dopamine, prompting us to compare ourselves with others (**social influence**)
 - In general, today’s technology is designed to keep us “engaged,” offering us a mix of old and new content each time
 - a mechanism that is used by **slot machines**, too

Attention-Capture Damaging Patterns: Impacts

- ACDPs may negatively affect people's digital well-being
 - They promote “**digital addiction**”
 - They undermine users' **attention** and **productivity**
 - They undermine users' **sense of agency** and self-control
 - They result in a later **sense of regret**

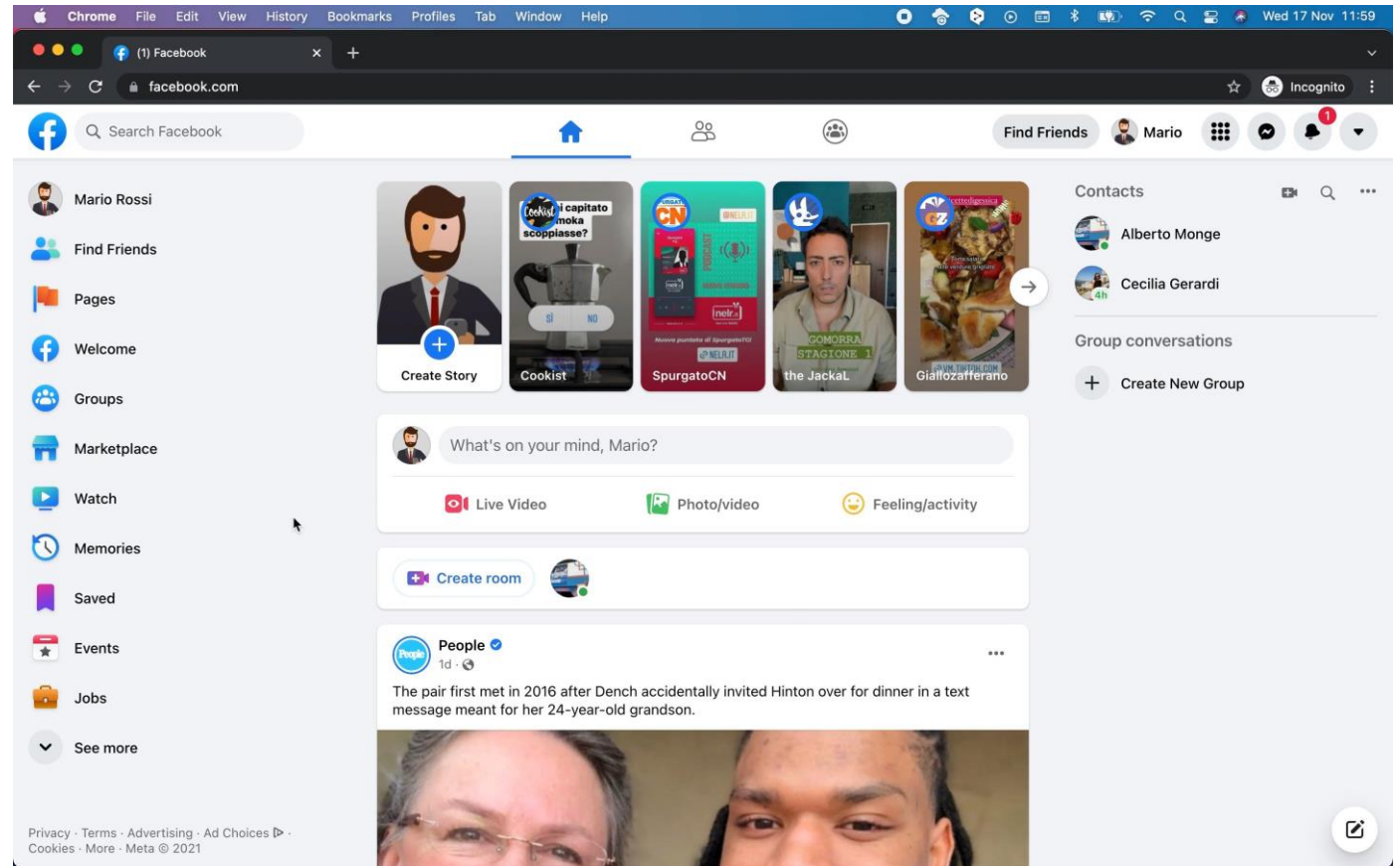
Attention-Capture Damaging Patterns: Examples

**GUILTY-PLEASURE
RECOMMENDATIONS**

NEVERENDING AUTOPLAY

CASINO PULL-TO-REFRESH

INFINITE SCROLLING



Attention-Capture Damaging Patterns: Examples

GUILTY-PLEASURE RECOMMENDATIONS

NEVERENDING AUTOPLAY

CASINO PULL-TO-REFRESH

INFINITE SCROLLING

- They are based on:
 - Previous user's interactions (**Content-Based**)
 - Preferences of similar users (**Collaborative Filtering**)
- Recommendations are useful if the goal of the platform matches the user's goals (**value-alignment problem**)
- They can become a «**trap**» for keeping the user's attention on the platform against the user's will
- They exploit **variable reward techniques**

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- A new video is **automatically played** when the current one finishes. There is never a point for the user to **stop** and **reflect**, and the option to turn off autoplay is hidden or non-existent
- Autoplay may be **useful** in some circumstances, e.g., listening to music on YouTube while working
- It can **prolong** usage sessions («*let me see this new video and then I will close the app!*»)
- It exploits **variable reward techniques**
- Reduces the **user's autonomy**

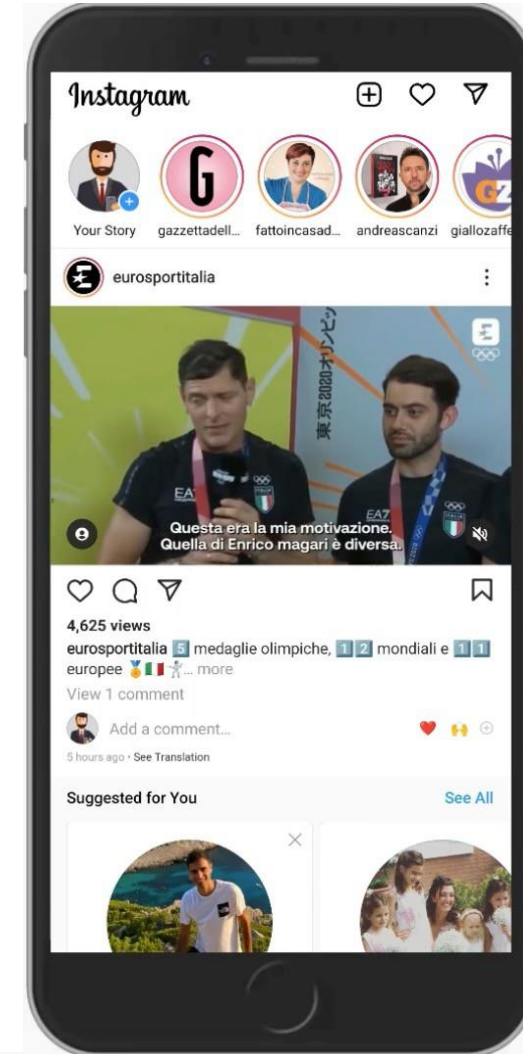
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- When the user **swipes down** on their smartphone, there is an **animated reload** of the page that **may or may not** reveal new appealing content.
- Users may be tempted to refresh compulsively, hoping to receive new content
- It exploits **variable reward techniques**
- Similar to **slot machines**

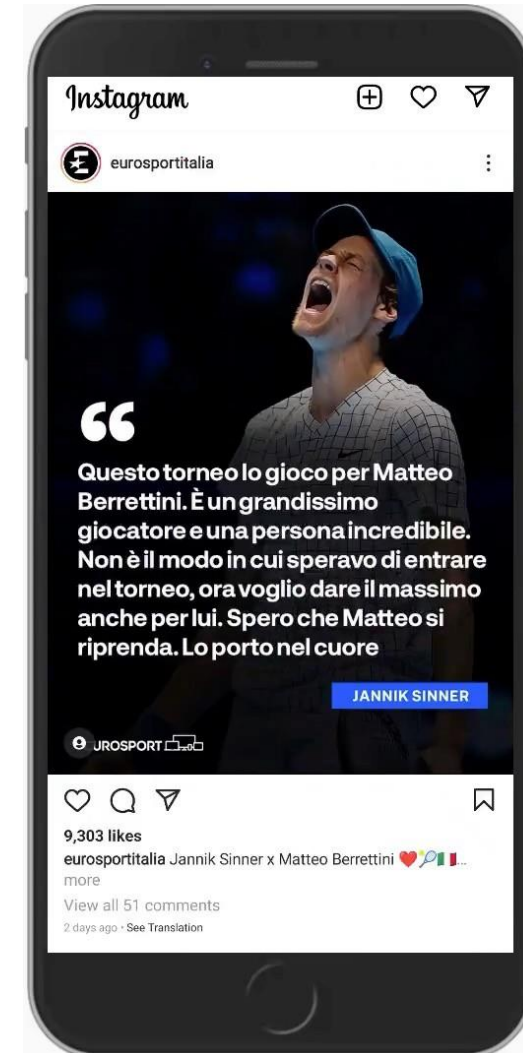
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- As the user scrolls down a page, more content **automatically** and **continuously loads** at the bottom.
- It **decreases** the **effort** required to browse content
- Promotes “**endless**” usage sessions
- It exploits **variable reward techniques**

References

■ Some material from

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